

Lillya Maria Mélo

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EDUCATION

State University of Paraíba

Bachelor's Degree in Computer Science

Campina Grande, PB

Jan. 2025 – Present

EXPERIENCE

Undergraduate Research Student

Jun. 2025 – Present

State University of Paraíba

Labmulti - CCT, PB

- Research project focused on developing Convolutional Neural Networks (CNNs) to automate the detection of water tanks using satellite imagery.
- Developed using Python, Ultralytics YOLOv8, OpenCV, LabelImg, PyTorch, and NumPy, including the creation of a custom dataset.
- End-to-end pipeline: data collection and annotation, preprocessing, data augmentation, dataset splitting, and model validation.
- The final goal is to deliver a functional prototype to assist in identifying and preventing dengue outbreak hotspots.
- The solution is planned to be adopted by multiple Municipal Health Departments across Brazil, with future integration into a web platform where I will contribute to backend development.

Software Development Intern

Jan. 2026 – Present

INOVAÍ - Laboratory

Piraí, Rio de Janeiro, Brazil

- Contributed to both product design and software development, participating in the creation of prototypes and the implementation of production-ready code for mobile and desktop platforms.
- Designed interactive prototypes to support the planning, validation, and usability evaluation of digital solutions.
- Contributed to the development of an internal service request and inventory management system using Python and Django.
- Implemented backend features including business logic, data modeling, database integration, and workflow support for operational processes.
- Worked on frontend development using HTML, CSS, and JavaScript to build responsive interfaces and improve user experience across web and mobile environments.

PROJECTS

Inventory Management and Registration System (Capstone Project) | *Python, Django* Nov. 2025 – Jan. 2026

- Backend development and integration of an inventory control system with expiration monitoring and automated alerts.
- Implemented customer, supplier, and product registration with search and filtering features.
- Developed full CRUD operations with business rules and a relational database.
- Built automated alert systems integrated with WhatsApp and Email notifications.

SKILLS

Programming Languages: Python, Java, JavaScript, C, SQL (Postgres, MySQL, SQLite), HTML/CSS

Frameworks: Django, Django REST, Swagger

Developer Tools: Git, Conda/Miniconda, VS Code, Jupyter Notebook, Twilio

Libraries: pandas, NumPy, Matplotlib, OpenCV, TensorFlow

EXTRACURRICULAR ACTIVITIES

- Participated in the Latin American Women's Programming Marathon (2025), ranking among the top 100 teams.

LANGUAGES

Portuguese : Native.

English: Advanced (Reading, Writing, Speaking)

Spanish:Intermediate (Reading, Writing, Speaking)